

47 Animal Development

KEY CONCEPTS

- 47.1** Fertilization and cleavage initiate embryonic development *p.* 1044
- 47.2** Morphogenesis in animals involves specific changes in cell shape, position, and survival *p.* 1049
- 47.3** Cytoplasmic determinants and inductive signals regulate cell fate *p.* 1057

Study Tip

Use word roots to learn vocabulary:

Word roots often provide clues to the meaning of biological terms. Study this list of word roots and their meanings to help you infer and remember the definitions of vocabulary words from this chapter and others.

Word root—meaning

blast—group of immature cells	poly—many	endo—internal
tropho—change	derm—tissue	meso—middle
coel—cavity	genesis—origin	ecto—external
morpho—shape	diplo—two	
	triplo—three	

Term—definition

blastocoel—cavity in ball of immature cells
diploblast
triploblast
trophoblast
endoderm
morphogenesis

Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 47
- Video: Sea Urchin Fertilization and Cleavage
- HHMI Video: Differentiation and the Fate of Cells

For Instructors to Assign (in Item Library)

- Tutorial: Visualizing Gastrulation
- Experimental Inquiry: How Do Calcium Ions Help to Prevent Polyspermy During Egg Fertilization?

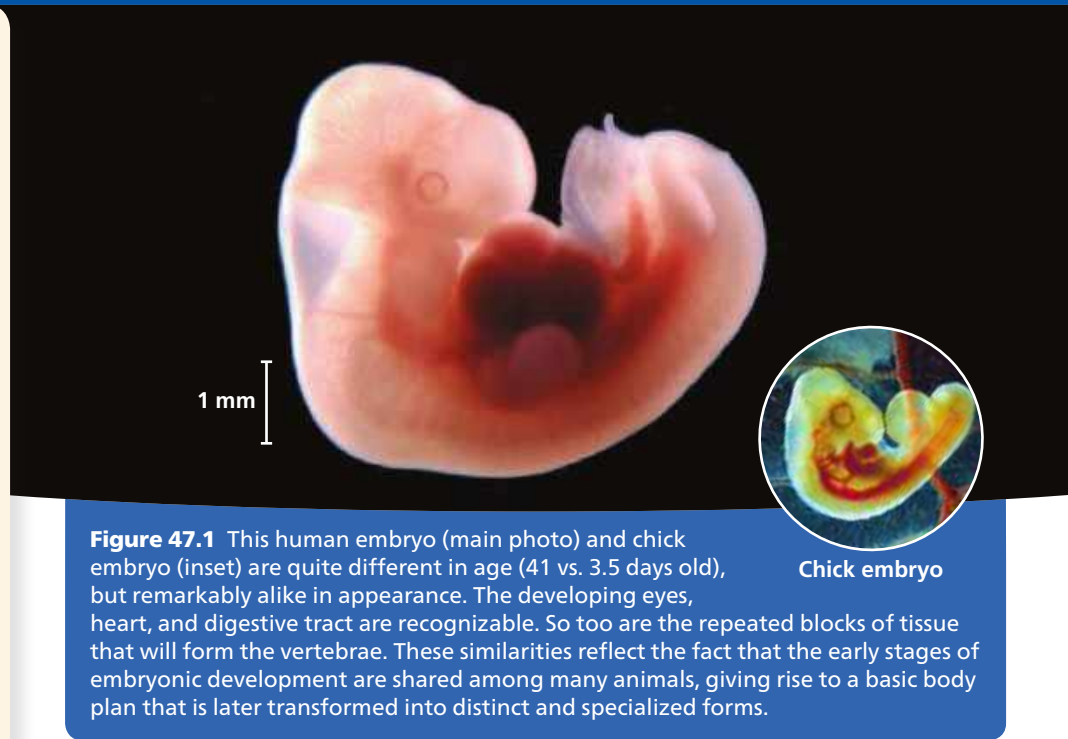


Figure 47.1 This human embryo (main photo) and chick embryo (inset) are quite different in age (41 vs. 3.5 days old), but remarkably alike in appearance. The developing eyes, heart, and digestive tract are recognizable. So too are the repeated blocks of tissue that will form the vertebrae. These similarities reflect the fact that the early stages of embryonic development are shared among many animals, giving rise to a basic body plan that is later transformed into distinct and specialized forms.

What processes transform an egg into an embryo with recognizable structures?

